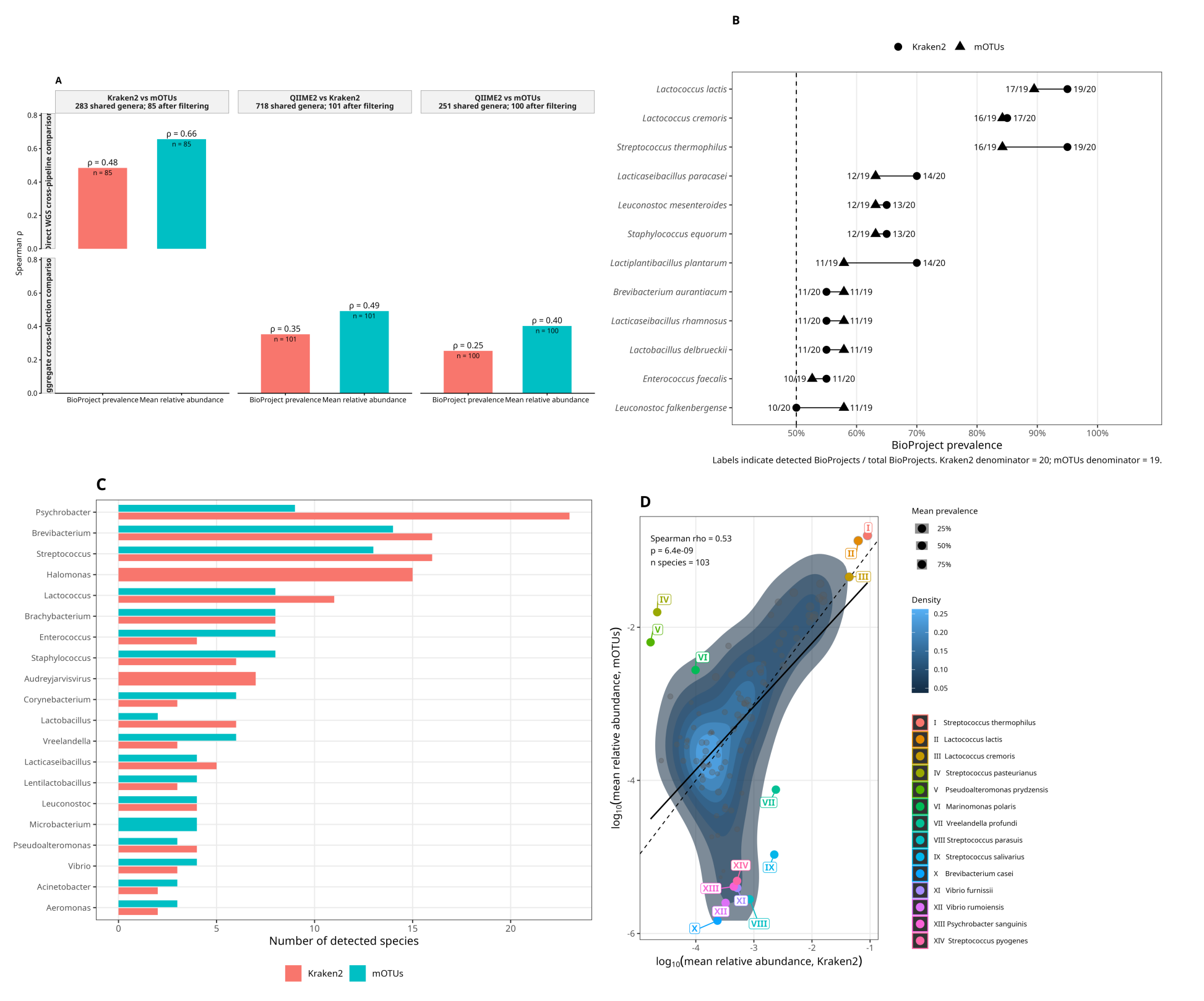

A
Kraken2 vs mOTUs
283 shared genera; 85 after filtering
QIIME2 vs Kraken2
718 shared genera; 101 after filtering
QIIME2 vs mOTUs
251 shared genera; 100 after filtering
Direct WGS cross-pipeline comparison
Aggregate cross-collection comparison
Spearman ρ
ρ = 0.48
n = 85
ρ = 0.66
n = 85
ρ = 0.35
n = 101
ρ = 0.49
n = 101
ρ = 0.25
n = 100
ρ = 0.40
n = 100
BioProject prevalence
Mean relative abundance
B
Kraken2
mOTUs
Lactococcus lactis
Lactococcus cremoris
Streptococcus thermophilus
Lacticaseibacillus paracasei
Leuconostoc mesenteroides
Staphylococcus equorum
Lactiplantibacillus plantarum
Brevibacterium aurantiacum
Lacticaseibacillus rhamnosus
Lactobacillus delbrueckii
Enterococcus faecalis
Leuconostoc falkenbergense
17/19
19/20
16/19
17/20
16/19
19/20
12/19
14/20
12/19
13/20
12/19
13/20
11/19
14/20
11/20
11/19
11/20
11/19
11/20
11/19
10/19
11/20
10/20
11/19
50%
60%
70%
80%
90%
100%
BioProject prevalence
Labels indicate detected BioProjects / total BioProjects. Kraken2 denominator = 20; mOTUs denominator = 19.
C
Psychrobacter
Brevibacterium
Streptococcus
Halomonas
Lactococcus
Brachybacterium
Enterococcus
Staphylococcus
Audreyjarvisvirus
Corynebacterium
Lactobacillus
Vreelandella
Lacticaseibacillus
Lentilactobacillus
Leuconostoc
Microbacterium
Pseudoalteromonas
Vibrio
Acinetobacter
Aeromonas
0
5
10
15
20
Number of detected species
Kraken2
mOTUs
D
Spearman rho = 0.53
p = 6.4e-09
n species = 103
log10(mean relative abundance, mOTUs)
log10(mean relative abundance, Kraken2)
-2
-4
-6
-4
-3
-2
-1
I
II
III
IV
V
VI
VII
VIII
IX
X
XI
XII
XIII
XIV
Mean prevalence
25%
50%
75%
Density
0.25
0.20
0.15
0.10
0.05
I Streptococcus thermophilus
II Lactococcus lactis
III Lactococcus cremoris
IV Streptococcus pasteurianus
V Pseudoalteromonas prydzensis
VI Marinomonas polaris
VII Vreelandella profundi
VIII Streptococcus parasuis
IX Streptococcus salivarius
X Brevibacterium casei
XI Vibrio furnissii
XII Vibrio rumoiensis
XIII Psychrobacter sanguinis
XIV Streptococcus pyogenes